

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : Cyber & Information Security

Aim: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

What is a Message Authentication Code (MAC)?

A Message Authentication Code (MAC) is a cryptographic value generated from a message and a secret key. It is used to verify both the integrity and authenticity of the message.

A MAC ensures that:

- The message has not been altered during transmission.
- The message was sent by someone who possesses the correct secret key.

Unlike a simple hash function, a MAC requires a secret key, making it much more secure for authentication purposes.

HMAC (Hash-based Message Authentication Code)

HMAC is one of the most commonly used MAC algorithms. It combines:

- A secret key
- A cryptographic hash function (SHA-256)

to produce a secure authentication code.

Formula

$$\text{HMAC} = \text{Hash}(\text{Secret Key} + \text{Message})$$

In this practical, the SHA-256 hashing algorithm is used with HMAC.

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Working Principle

1. The sender enters a secret key.
2. The sender enters the message.
3. The system generates a MAC using HMAC-SHA256.
4. The sender transmits the message along with the generated MAC.
5. The receiver receives both the message and MAC.
6. Using the same secret key, the receiver generates a new MAC.
7. The generated MAC is compared with the received MAC.
8. If both MACs match, the message is authentic and has not been modified.
9. Otherwise, the message has been tampered with.

Algorithm: HMAC-SHA256 for Message Authentication Code (MAC)

Algorithm Name

HMAC-SHA256 (Hash-based Message Authentication Code using SHA-256)

Input

- Secret Key (**K**)
- Message (**M**)

Output

- Message Authentication Code (MAC)
- Verification result (Valid / Invalid)

Steps

1. Start the program.
2. Read the secret key **K** from the user.
3. Read the message **M** from the user.
4. Convert both the secret key and message into byte format.
5. Apply the **HMAC-SHA256** algorithm using the secret key and the message.
6. Generate the MAC (authentication code).

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7. Display the generated MAC.
8. Read the received message and the received MAC.
9. Again compute the MAC using the received message and the same secret key.
10. Compare the newly generated MAC with the received MAC using a secure comparison method (`compare_digest()`).
11. If both MAC values are identical:
 - Display **"MAC Verified Successfully."**
 - Display **"Message Integrity Maintained."**
 - Display **"Sender Authenticated."**
12. Otherwise:
 - Display **"MAC Verification Failed."**
 - Display **"Message has been Tampered."**
13. Stop the program.

Python Program

```
import hmac

import hashlib

# Secret key shared between sender and receiver
secret_key = input("Enter Secret Key: ").encode()

# Original message
message = input("Enter Message: ").encode()

# Generate MAC using HMAC-SHA256
mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:")

print(mac)
```

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```
print("\n===== Verification =====")

# Receiver enters received message
received_message = input("Enter Received Message: ").encode()

# Receiver enters received MAC
received_mac = input("Enter Received MAC: ")

# Generate MAC again
new_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()

# Verify MAC
if hmac.compare_digest(new_mac, received_mac):
    print("\n MAC Verified Successfully")
    print(" Message is Authentic")
    print(" Data Integrity Maintained")
else:
    print("\n Verification Failed")
    print(" Message has been Tampered")
```

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Output

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/tycs_103/cyber/mac_prac.py =====
Enter Secret Key: 12886213s
Enter Message: soham patil

Generated MAC:
e4bfd69389e34a83015ccdc5f242664016bdd674aa3388a2cd624f0a3bd289

===== Verification =====
Enter Received Message: soham patil
Enter Received MAC: e4bfd69389e34a83015ccdc5f242664016bdd674aa3388a2cd624f0a3bd289

MAC Verified Successfully
Message is Authentic
Data Integrity Maintained

>>>
```

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/tycs_103/cyber/mac_prac.py =====
Enter Secret Key: 1234670s1d2
Enter Message: soham patil

Generated MAC:
07b239a3aed94ca8fecaf0a4faaab80139574f7f658ae7840cb158612cbd5af9

===== Verification =====
Enter Received Message: patil soham
Enter Received MAC: 07b239a3aed94ca8fecaf0a4faaab80139574f7f658ae7840cb158612cbd5af9

Verification Failed
Message has been Tampered

>>>
```